

Umbraco LFI Exploitation since 2015

Umbraco LFI Exploitation since 2015 - Boik Su, @boik_su, Medium.

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Umbraco

Local File Inclusion

Lfi

Aspnet

Rce

Umbraco LFI Exploitation since 2015

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[image: Boik Su]

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Boik Su

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Share

Umbraco CMS

Umbraco is the leading Open Source ASP.NET CMS and currently, more than 500,000 websites worldwide are powered by the flexible and editor-friendly CMS.

TL;DR

In a nutshell, the ClientDependency library whose version is from **v1.8.2.1** to ****v1.9.8**** is affected and prone to LFI.

Umbraco includes a **ClientDependency** handler that is accessed as follows:

```
http://localhost/DependencyHandler.axd?
s=ZmlsZTovL2xvY2FsaG9zdC9jJC9XaW5kb3dzL3dpbi5pbmk=&t=3&cdv=1
```

, where **s** ****is the parameter whose value is a **base64-encoded** string, normally referencing local file paths of client-side resources to minify or concatenate at run-time (e.g. `/assets/foo.js`). The Umbraco framework will reference the handler instead of **.css** and **.js** files directly, as the handler returns the processed file contents in its HTTP response.

In the above case, the **s** parameter is `[http://localhost/packages.config#.css]` (`http://localhost/web.config#.css`) encoded, which allows the private file to be read and displayed from the HTTP response:

Content of `[http://localhost/packages.config#.css]` (`http://localhost/web.config#.css`)

Besides, if we can read the `machineKey` from `web.config`, we can even achieve pre-auth RCE on **Umbraco 7 and below!**

Background

The story behind this vulnerability is, I was testing this [vulnerability](#) at the moment. Since there's no public information about this vulnerability, I started searching for old vulns pertaining to [ClientDependency](#). Luckily, there are two write-ups ([1](#), [2](#)) talking about similar vulnerabilities reported before.

Coincidentally, I was practicing to use CodeQL to model vulns and trying to do the Variant Analysis. It seems to be a good target for me because this library has suffered from the LFI issue several times.

If you're not familiar with CodeQL, you might want to check out the [story](#) that I published before. 😊

The culprit

In order to model the flaw with dynamic/taint analysis, two things you should have for CodeQL are the source and sink nodes. As the two write-ups described, the source is no doubt the user-controllable input, specifically the `s` parameter; while sinks remain unknown to us. However, it's easy to find out `WriteFileToStream` is possibly the function responsible for reading out the file's content to the HTTP response after reviewing the source code.

Let's try to write a simple query to identify possible flows from a user-controllable input to any arguments of the mentioned function:

```
**import** csharp
**import** DataFlow
**import** DataFlow::PathGraph
**import** semmle.code.csharp.security.dataflow.flowsources.Remote**class**
SuspiciousFlow **extends** TaintTracking::Configuration {
    SuspiciousFlow() { this = "SuspiciousFlow" } **override** **predicate** isSource(Node
source) {
    source **instanceof** AspNetQueryStringRemoteFlowSource
** override** **predicate** isSink(Node sink) {
    **exists**(MethodCall mc |
        mc.getTarget().getName() = " WriteFileToStream " **and**
        sink = exprNode(mc.getAnArgument())
    )
}
**from** SuspiciousFlow s, PathNode source, PathNode sink
**where** s.hasFlowPath(source, sink)
**select** sink.getNode(), source, sink, "This input flows to $@.", sink.getNode(),
"here"
```

Apparently, there are some dead paths preventing the flows, and we need to fix them by allowing more tainted steps. To do so, we need to know how this library normally works.

Right now, let's get the developing environment all set, and see how and what this library loads the files.

The normal processing flow of the library

The series of screenshots shown below are taken with version **v1.9.6**.

It all starts from here. A user gives the endpoint the **s** parameter, denoting the file about to be read.

Load "/umbraco/assets/css/umbraco.css"

Then, the value of **s** is base64-decoded and processed as follows.

Step 1, decode the filekey and validate the type:

Step 2, the `WritePathToStream` function is responsible for checking the file and writing its content to the stream:

Step 3, extract the **extension** from the path and make sure if it's one of the **registered extensions (default: .css, .js)**:

The `CanProcessLocally` function is responsible for checking if the file is accessible locally:

"/umbraco/assets/css/umbraco.css" is accessible locally

Step 4, if the file exists, locate, compress, and write it back to the stream:

Locate and load the file if existing

Prepare to compress the content

At this moment, the file is fetched locally and will be sent back as the HTTP Response body. Everythings are done here.

We used the ****local relative path**** in the above case. This time, we use ****URI**** instead.

As you can imagine, the `CanProcessLocally` function throws an exception:

Therefore, the process will jump into the catch block:

In this block, the process is trying to load the file locally again if it's actually a ****local relative request**** before issuing the external request:

"uri.Scheme" should be either "http" or "https"

"http.Request.Url.Host" equals to "uri.Host" means it's a local request

If the file exists locally, read the file and return its content just like the first case, or issue the external request.

To finalize...

So, after getting the developing environment all set and knowing what blocks the flows from the source to the sinks possibly, we're able to modify the query to a more general one.

```

**import** csharp
**import** DataFlow
**import** DataFlow::PathGraph **class** WebIHttpHandler **extends** Method {
  WebIHttpHandler() {
    getDeclaringType().
      getABaseType().hasQualifiedName("System.Web.IHttpHandler") **and**
      isSourceDeclaration()
  }
} **class** SuspiciousFlow **extends** TaintTracking::Configuration {
  SuspiciousFlow() { this = "SuspiciousFlow" } **override** **predicate** isSource(Node
source) {
  **exists**(WebIHttpHandler web, MethodCall mc |
    localExprFlow(web.getAParamenter().getAnAccess(), mc.getAnArgument()) **and**
    source = exprNode(web.getAParameter().getAnAccess())
  }
} **override** **predicate** isSink(Node sink) {
  **exists**(MethodCall mc |
    mc.getTargetVariable().getName() = "WriteFileToStream" **and**
    sink = exprNode(mc.getAnArgument())
  }
} **override** **predicate** isAdditionalTaintStep(Node src, Node dst) {
  **exists**(MethodCall mc, Type type |
    **if** mc.getTarget() **instanceof** ExtensionMethod
    **then** (
      **if** mc.getTarget().getName() = "Select"
      then (
        (
          src = exprNode(mc.getArgument(0))
          **and**
          dst = exprNode(mc.getArgument(1).(LambdaExpr).getParamenter(0).getAnAccess())
        )
        **or**
        (
          src = exprNode(mc.getArgument(1).(LambdaExpr).getExpressionBody())
          **and**
          dst = exprNode(mc)
        )
      ) **else** (
        type = mc.getTarget().getReturnType()
        **and**
        **not** type **instanceof** VoidType
        **and**
        src = exprNode(mc.getArgument(0))
        **and**

```

```

        dst = exprNode(mc)
    )
) **else** (
    type = mc.getTarget().getReturnType()
    **and**
    **not** type **instanceof** VoidType
    **and**
    mc.getChildExpr(-1) **instanceof** MethodCall
    **and**
    src = exprNode(mc.getChildExpr(-1))
    **and**
    dst = exprNode(mc)
)
)
}
}**from** SuspiciousFlow s, PathNode source, PathNode sink
**where** s.hasFlowPath(source, sink)
**select** sink.getNode(), source, sink, "This input flows to $@.", sink.getNode(),
"here"

```

We're finally able to model the same vulnerability with CodeQL and let's apply it to the latest version of the library to see if we can find something different.

Interestingly, the same path still exists against the latest version of it (v1.9.8). I believe that it's because I don't add any sanitizers there, but some functions' outputs should block the flow to `WriteFileToStream`.

Anyways, I can confirm now that this vulnerable path is still there as long as I can find a way to bypass the patch of the previous flaw, and after a little time, I unveil a key point that allows me to bridge the gap between two nodes and make the path accessible one more time. □

Analysis

The bug is residing in the [handling part](#) of processing external requests, which is:

```

**if** (ex **is** NotSupportedException
    || ex **is** ArgumentException
    || ex **is** HttpException)
{
    //could not parse the string into a fileinfo or couldn't mappath, so we assume it is
    a URI
    //before we try to load it by URI, we want to check if the URI is a local request,
    we'll try to detect if it is and
    // then try to load it from the file system, if the file isn't there then we'll
    continue trying to load it via the URI.
    Uri uri;
    **if** (Uri.TryCreate(path, UriKind.RelativeOrAbsolute, out uri) &&
uri.IsLocalUri(context))
    {
        **var** localPath = uri.PathAndQuery;
        **var** fi = new FileInfo(context.Server.MapPath(localPath));
        **if** (fi.Exists)
        {
            **try**
            {
                WriteFileToStream(sw, fi, type, path, context); //internal request
            }
            **catch** (Exception ex1)
            {
                ClientDependencySettings.Instance.Logger.Error($"Could not load file
contents from {path}. EXCEPTION: {ex1.Message}", ex1);
            }
        }
    }
    def = WriteFileToStream(sw, path, type, context);
}

```

The old attack vector `http://localhost/web.config` won't work because of the validation of the extension, which is **.config** here. However, as you can see, if the following two conditions `(Uri.TryCreate(path, UriKind.RelativeOrAbsolute, out uri) and uri.IsLocalUri(context))` both return true, the `localPath` next line will be `uri.PathAndQuery`.

What if we passed `[http://localhost/web.config#.css](http://localhost/web.config#.css)`? The `uri.PathAndQuery` equals to `/web.config`, and therefore, we can successfully fetch the content of `context.Server.MapPath(localPath)`, which is `C:\[WEBSITE]\web.config`. 🤖🤖🤖

After checking the logic behind it, the flaw could also apply to the following versions:
v1.8.2.1 to v1.9.8.

To leak machineKey possibly: `/web.config, /bin/[APP_NAME].dll.config`

To leak the full application path possibly: `/App_Data/umbraco.licensing.log.txt`

Exploitation

If we get the `machineKey` from the `web.config`, we can then force the Umbraco CMS to deserialize malign objects and lead to the RCE. However, it depends on what version of Umbraco you encounter.

As of **Umbraco v8**, it officially abandoned the support of WebForm, and therefore, it will not be vulnerable to deserialization issues caused by machineKey. You can still try to read other files that possibly leak other sensitive information, though.

However, if you encountered a website running with **Umbraco v4 to v7**; congrats, you can directly trigger the deserialization on **Login Page or Ping Page**. ☐☐☐

whoami

Umbraco7_RCE.js

[

umbraco7_exp_with_no_creds.js - dotnet_research - Visual Studio Code 2020-07-23 18-18-53.mp4

Umbraco 7 RCE video

drive.google.com

](https://drive.google.com/file/d/1DePUWZigtmQ8XvBUonjrn8k5ilyFGY-I/view?usp=sharing&source=post_page-----d32803661fa3-----)

Patch

This time, Umbraco decides to re-check the file's extension before it tries to read the content out, and this patch won't let attackers exploit similar issues once again!

The re-validation of the file's extension in WriteFileToStream

Indicators of Compromise

Since this attack would make a call to ClientDependency, the library will create cached files for references. These cached files store **evidence of the possible breach**.

For most people, the files are stored in `~/App_Data/Temp/ClientDependency`, but it is possible that they are stored somewhere else (on Azure specifically). Make sure to verify the location, examples on how to do that are available in the forum post about this security update: <https://our.umbraco.com/forum/using-umbraco-and-getting-started/101529-security-update-for-march-2020>

Q & A

- Why can't I include a file sometimes while disguising it as a **Css** or a **Js** file?

~~It's because that starting from ClientDependency v.1.9.0, the minifying process of Css/Js has changed by introducing customized minifiers, which don't allow a line feed to be contained inside a string literal. That is, files containing this kind of pattern would break the minifiers!~~ You can! You can bypass the limitation by setting variable **t** to an integer ≥ 3 , and this is the expected behavior of the .NET **`**Enum.Parse**`** function. **When you provide an integer to `**Enum.Parse`**, the function will try to get the proper Enum value. However, if the index is out of range it will return an undefined Enum. This will ensure you skip the minification process and get the content of any file you want.

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